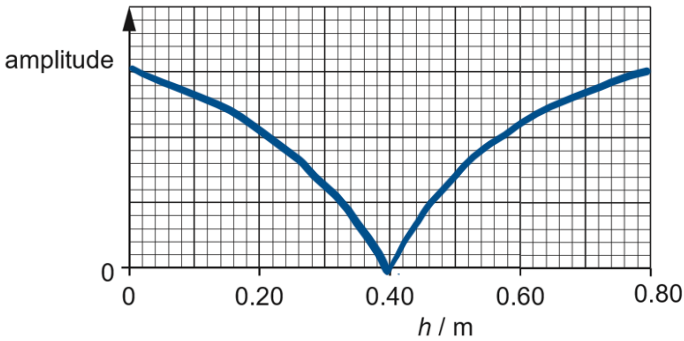


6a	distance moved by wavefront/energy during one cycle/oscillation/period (of source)	B1
6b	$T = 2.0 \times 2.5$ $(= 5.0 \text{ ms})$ $f = 1 / (5.0 \times 10^{-3})$ $= 200 \text{ Hz}$	A1
6c(i)	(incident) wave reflects at end/top of tube (incident) wave and reflected wave interfere/superpose to produce stationary wave	B1 B1
6c(ii)	line has maximum value of amplitude at $h = 0$ and $h = 0.80 \text{ m}$ only line has minimum/zero value of amplitude at $h = 0.40 \text{ m}$ only 	B1 B1
6c(iii)1.	vertical/along length of tube/along axis of tube	B1
6c(iii)2.	phase difference = 0°	A1
6c(iv)	$\lambda = 2 \times 0.80 = 1.6 \text{ m}$	A1
6d(i)	path difference = $8.0 + (20.8^2 - 8.0^2)^{0.5} - 20.8$ $= 6.4 \text{ m}$	M1 A0
6d(ii)	path difference = 4λ waves (meet at C) in phase constructive interference (of waves) occurs	B1 B1
6d(iii)	$v = 200 \times 1.6$ $= 320 \text{ m s}^{-1}$ $\Delta t = 6.4 / 320$ or $27.2 / 320 - 20.8 / 320$ $= 0.020 \text{ s}$	C1 A1
6d(iv)	$3\lambda = 6.4$	C1

	$\lambda = 2.1 \text{ m}$	A1
6e(i)	Use Malus' law, Intensity $I = I_0 \cos^2 \theta$ Intensity $I = I_0 \cos^2 \theta = I_0 \cos^2 45^\circ$ $= I_0/2$ And orientation to the vertical is 45°	C1 A1
6e(ii)	Intensity $I = (I_0 \cos^2 45^\circ) \cos^2 45^\circ$ $= I_0/4$ and orientation to the vertical is 0° (i.e. emergent beam is polarised vertically)	C1 A1
	Total:	20

7a(i)	Faraday's Law of Electromagnetic Induction states that the induced e.m.f is	P1
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